ALTAWAKOL
CABLES

TYPE-TESTED CONDUCTOR

ACSR 380/50 mm²

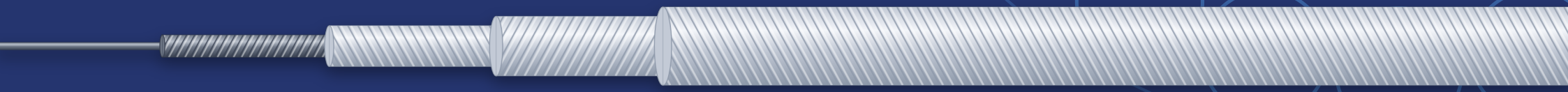
Type Test Results

نتائج الاختبارات النوعية

17 /17

type tests passed

Report No. 568/2026 · 29 September 2026
Extra High Voltage Research Center (EHVRC)



IEC 61089 · IEC 62641 · IEC 63248 · BS EN 50182 · BS EN 50189 · EDMS 04-100-1

THE PRODUCT

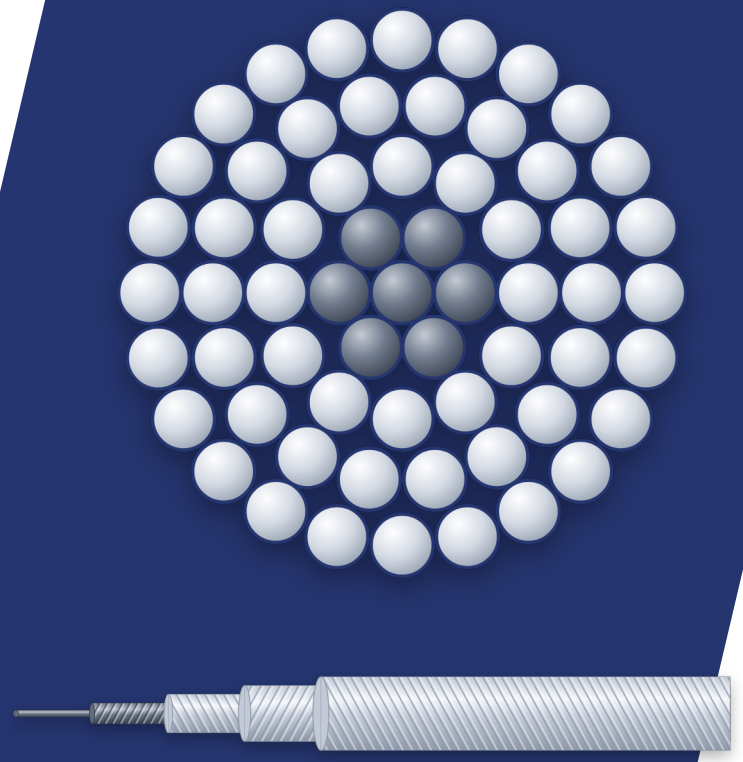
ACSR 380/50 at a Glance

مواصفات المنتج

A 54/7 aluminium conductor steel-reinforced: three layers of hard-drawn aluminium wires (12 + 18 + 24) stranded over a seven-wire galvanized steel core. Widely used on 220 kV and 500 kV transmission lines.

Designation	**380/50 (54/7)**
Aluminium area	**381.7 mm²**
Steel area	**49.5 mm²**
Total area	**431.2 mm²**
Aluminium wires	**54 × 3.00 mm (AL1)**
Steel wires	**7 × 3.00 mm (ST1A)**
Overall diameter	**27.0 mm**
Mass	**1442.5 kg/km**
DC resistance at 20 °C	**0.0758 Ω/km**
Rated tensile strength	**121.30 kN**
Current rating*	**820 A**
Product code	**ATC-ACS-012**

* Calculated per IEEE 738: ambient 35 °C, conductor 80 °C, wind 0.6 m/s, solar 900 W/m². For guidance only.



Steel core 7 × 3.00 mm · Aluminium layers 12 / 18 / 24 × 3.00 mm
Alternate lay directions · right-hand outer layer

INDEPENDENT VERIFICATION

Test Programme

برنامج الاختبارات

A sample of our ACSR 380/50 conductor was type-tested by the Extra High Voltage Research Center (EHVRC) — Laboratories, Research & Testing Sector — Egyptian Electricity Holding Company. All 17 tests met the requirements of the applicable IEC, BS EN and EDMS standards.

01	Surface condition	Construction	✓ PASS
02	Inertness	Construction	✓ PASS
03	Wrapping test — aluminium wires	Mechanical	✓ PASS
04	Wrapping test — zinc coating	Zinc coating	✓ PASS
05	Design & construction	Construction	✓ PASS
06	Direction of lay	Construction	✓ PASS
07	Lay ratio — outer layer	Construction	✓ PASS
08	Resistivity of Al wires at 20 °C	Electrical	✓ PASS
09	Mass of zinc coating	Zinc coating	✓ PASS
10	Mass per unit length	Construction	✓ PASS
11	Zinc dip test	Zinc coating	✓ PASS
12	Adhesion of zinc coating	Zinc coating	✓ PASS
13	Tensile strength of wires	Mechanical	✓ PASS
14	Elongation — steel wires	Mechanical	✓ PASS
15	Stress at 1 % extension	Mechanical	✓ PASS
16	Tensile test of the conductor	Mechanical	✓ PASS
17	Joints in aluminium wires	Mechanical	✓ PASS

Test details

Report No. 568/2026
Issued 29 September 2026
Sample received 14 Sep 2026
Testing 14 – 24 Sep 2026
Scope Type tests to IEC

Standards

IEC 61089
Round wire concentric lay overhead electrical stranded conductors

IEC 62641 : 2022
Aluminium and aluminium alloy wires for stranded conductors

IEC 63248 : 2022
Coated or cladded metallic wire for stranded conductors

BS EN 50182
Round wire concentric lay stranded conductors

BS EN 50189
Zinc-coated steel wires

EDMS 04-100-1 : 2020
Egyptian distribution specification — ACSR conductors

RESULTS · 1

Construction & Dimensions

التصميم والأبعاد

Design and construction requirements

Item	Unit	Requirement	Measured	Verdict
Overall conductor diameter	mm	27	27.09	✓ PASS
Diameter of aluminium wires	mm	3.00	3.01	✓ PASS
Number of aluminium wires	—	54	54	✓ PASS
Diameter of steel wires	mm	3.00	3.00	✓ PASS
Number of steel wires	—	7	7	✓ PASS

Tolerances: conductor diameter ± 1 %, aluminium wire diameter ± 1 %, steel wire diameter ± 0.05 mm · IEC 61089 §6.6.1–6.6.2, IEC 62641 Table 2, IEC 63248 Table A.4, EDMS 04-100-1

Surface condition

Free from nicks, indentations and other visible imperfections.

✓ PASS

Inertness

Wire ends stay in position when the conductor is cut.

✓ PASS

Direction of lay

Outer layer: required right-hand, determined right-hand.

✓ PASS

Measured value within the permitted band

Overall diameter — measured **27.09 mm**



Lay ratio — outer layer — measured **12.66**



Mass per unit length — measured **1458.1 kg/km**



permitted range · measured · nominal

RESULTS · 2

Electrical & Zinc Coating

الخواص الكهربائية والجلفنة

0.027869

Ω·mm²/m — resistivity of Al wires at 20 °C

limit ≤ 0.028264

61.9 %

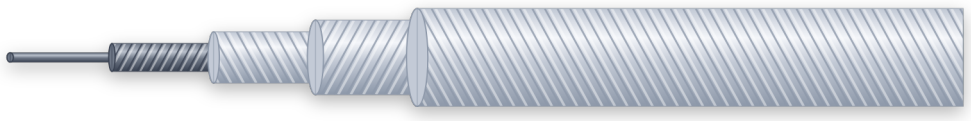
IACS conductivity equivalent

1.4 % better than the IEC limit

Resistivity of aluminium wires (Ω·mm²/m) measured **0.027869**

max 0.028264

Lower resistivity means lower DC resistance and lower line losses for the same aluminium area. Tested to IEC 62641 §6.4.7, Table 1.



ZINC-COATED STEEL CORE (ST1A) · ALUMINIUM LAYERS (AL1)

Zinc coating of steel core

269.3 g/m² · requirement ≥ 245

Coating mass 10 % above the class A minimum, for long-term corrosion protection of the steel core.

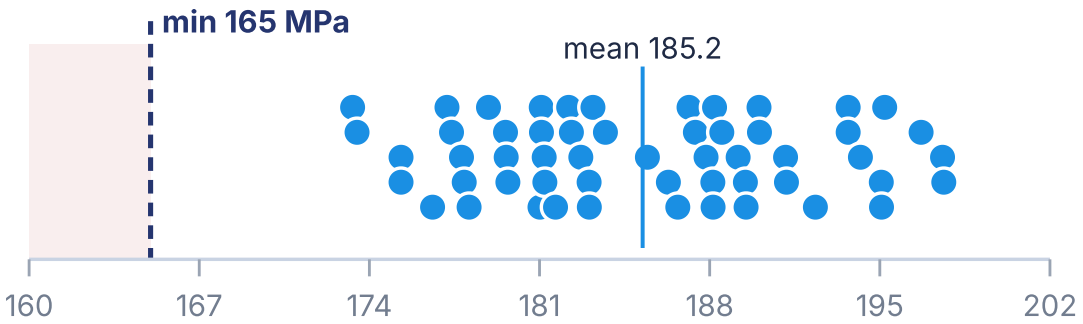
Zinc dip test (class A)	✓ PASS
Adhesion of zinc coating	✓ PASS
Wrapping test — zinc coating	✓ PASS
Wrapping test — aluminium wires	✓ PASS

RESULTS · 3

Mechanical Performance

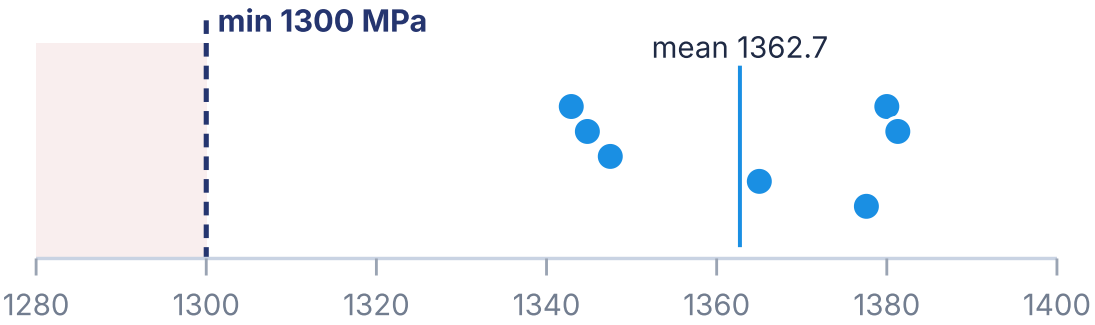
الخواص الميكانيكية

Tensile strength of aluminium wires — all 54 wires tested



Range 173.3 – 197.6 MPa · every wire above the 165 MPa minimum · IEC 62641 §6.4.3

Tensile strength of steel wires — all 7 wires tested

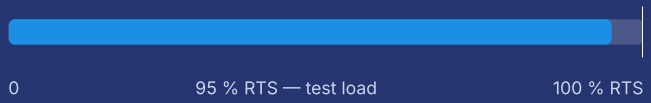


Range 1342.9 – 1381.3 MPa · every wire above the 1300 MPa minimum · IEC 63248 §7.4.3.2

TENSILE TEST OF THE CONDUCTOR

115.23 kN held 60 s

95 % of the rated tensile strength (RTS = 121.3 kN) applied to the complete conductor and held for 60 seconds — **no wire failure**. IEC 61089 §6.5.3 · EDMS 04-100-1.



Elongation — steel wires	≥ 3.5 %	**7.09 %**
Stress at 1 % extension	≥ 1100 MPa	**1241 MPa**
Joints in Al wires (cold-pressure butt weld)	≥ 130 MPa	**138.3 MPa**
Wrapping — aluminium wires	No break	**No break**

SUMMARY

Type Test Summary — ACSR 380/50 mm²

Report No. 568/2026 · Extra High Voltage Research Center (EHVRC) · issued 29 September 2026

ملخص نتائج الاختبارات

#	Test	Standard / clause	Requirement	Result	Verdict
01	**Surface condition**	BS EN 50182 §6.4.1 · IEC 61089 §6.6.5	Free from visible imperfections	**Pass**	✓ PASS
02	**Inertness**	BS EN 50182 §6.4.3	Wires stay in position when cut	**Pass**	✓ PASS
03	**Wrapping test — aluminium wires**	BS EN 50182 §6.5.2 · IEC 62641 §6.4.5	No break	**Pass**	✓ PASS
04	**Wrapping test — zinc coating**	BS EN 50182 §6.5.2 · IEC 63248 §7.4.4.2	No break	**Pass**	✓ PASS
05	**Design & construction**	IEC 61089 §6.6.1/6.6.2 · EDMS 04-100-1	Ø 27 mm · 54 × 3.00 Al · 7 × 3.00 St	**Ø 27.09 mm · 54 × 3.01 · 7 × 3.00**	✓ PASS
06	**Direction of lay**	IEC 61089 §5.4.2 · BS EN 50182 §6.4.4	Right-hand outer layer	**Right-hand**	✓ PASS
07	**Lay ratio — outer layer**	IEC 61089 §5.4.5 · BS EN 50182 §6.4.4	10 – 14	**12.66**	✓ PASS
08	**Resistivity of Al wires at 20 °C**	IEC 62641 §6.4.7, Table 1	≤ 0.028264 Ω·mm²/m	**0.027869 Ω·mm²/m**	✓ PASS
09	**Mass of zinc coating**	IEC 61089 §6.6.3 · BS EN 50182 §6.4.6	≥ 245 g/m²	**269.3 g/m²**	✓ PASS
10	**Mass per unit length**	IEC 61089 §6.6.3 · BS EN 50182 §6.4.6	1442.5 kg/km ± 2 %	**1458.1 kg/km**	✓ PASS
11	**Zinc dip test**	BS EN 50182 §6.5.2 · IEC 63248 §7.4.5.3	Class A coating	**Pass**	✓ PASS
12	**Adhesion of zinc coating**	BS EN 50189 §11.7 · IEC 63248 §7.4.5.2.1	Firmly adhered, no cracks or flaking	**Pass**	✓ PASS
13	**Tensile strength of wires**	IEC 63248 §7.4.3.2 · IEC 61089 §6.5	Al ≥ 165 MPa · St ≥ 1300 MPa	**Al 173–198 · St 1343–1381 MPa**	✓ PASS
14	**Elongation — steel wires**	BS EN 50182 §6.5.2 · IEC 63248 §7.4.3	≥ 3.5 %	**7.09 %**	✓ PASS
15	**Stress at 1 % extension**	BS EN 50182 §6.5.2 · IEC 63248 §7.4.3.2	≥ 1100 MPa	**1241 MPa**	✓ PASS
16	**Tensile test of the conductor**	IEC 61089 §6.5.3 · EDMS 04-100-1	Hold 115.23 kN (95 % RTS) for 60 s	**No failure**	✓ PASS
17	**Joints in aluminium wires**	IEC 61089 §6.5.4	Withstand 130 MPa	**Failed at 138.3 MPa**	✓ PASS

Conclusion of the laboratory: the ACSR 380/50 mm² conductor achieved the requirements of the tests according to IEC 61089, IEC 63248, BS EN 50182, BS EN 50189, IEC 62641 and EDMS 04-100-1.

Results apply to the specimen tested. The complete signed test report is reproduced in full on the following pages.

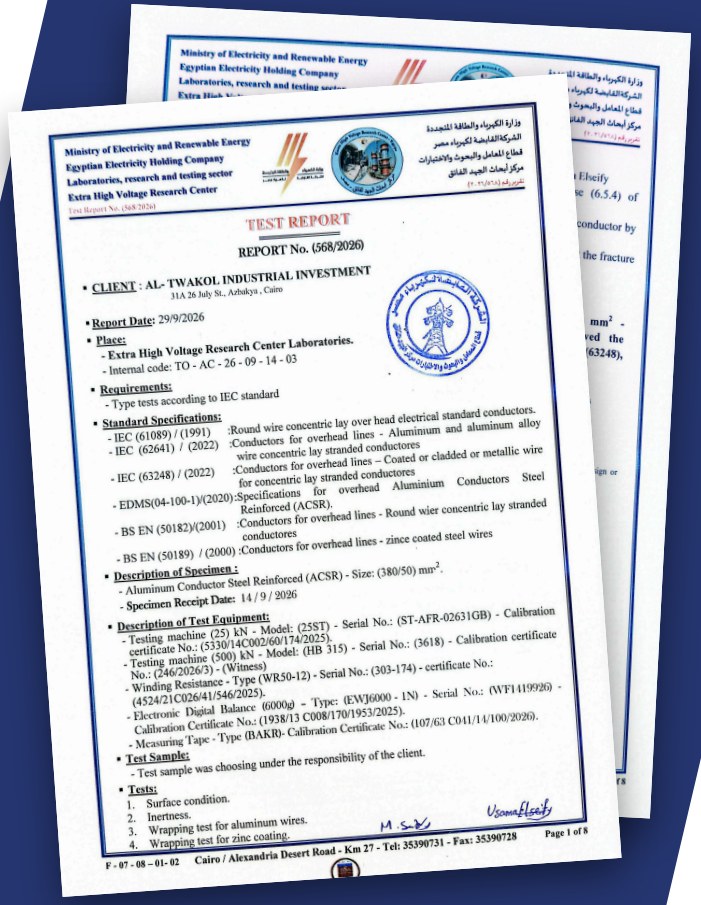
APPENDIX

Test Report & Certificates

شهادات الاختبار

The following pages reproduce, in full and unaltered, the signed and stamped test report issued by the laboratory for the ACSR 380/50 mm² conductor.

Report number	**568/2026**
Date of issue	**29 September 2026**
Laboratory	**EHVRC — Egyptian Electricity Holding Co.**
Specimen	**ACSR 380/50 mm²**
Tests	**17 type tests — all passed**
Pages	**8 of 8**


Ministry of Electricity and Renewable Energy
Egyptian Electricity Holding Company
Laboratories, research and testing sector
Extra High Voltage Research Center

TEST REPORT

REPORT No. (568/2026)

- CLIENT : AL- TWAKOL INDUSTRIAL INVESTMENT
  31A 26 July St., Azbakya , Cairo
- Report Date: 29/9/2026
- Place:
  - Extra High Voltage Research Center Laboratories.
  - Internal code: TO - AC - 26 - 09 - 14 - 03
- Requirements:
  - Type tests according to IEC standard
- Standard Specifications:
  - IEC (61089) / (1991) :Round wire concentric lay over head electrical standard conductors.
  - IEC (62641) / (2022) :Conductors for overhead lines - Aluminium and aluminum alloy wire concentric lay stranded conductores
  - IEC (63248) / (2022) :Conductors for overhead lines – Coated or cladded or metallic wire for concentric lay stranded conductores
  - EDMS(04-100-1)(2020):Specifications for overhead Aluminium Conductors Steel Reinforced (ACSR).
  - BS EN (50182)/(2001) :Conductors for overhead lines - Round wier concentric lay stranded conductores
  - BS EN (50189) / (2000) :Conductors for overhead lines - zince coated steel wires
- Description of Specimen :
  - Aluminum Conductor Steel Reinforced (ACSR) - Size: (380/50) mm².
  - Specimen Receipt Date: 14 / 9 / 2026
- Description of Test Equipment:
  - Testing machine (25) kN - Model: (25ST) - Serial No.: (ST-AFR-02631GB) - Calibration certificate No.: (5330/14C002/60/174/2025).
  - Testing machine (500) kN - Model: (HB 315) - Serial No.: (3618) - Calibration certificate No.: (246/2026/3) - (Witness)
  - Winding Resistance - Type (WR50-12) - Serial No.: (303-174) - certificate No.: (4524/21C026/41/546/2025).
  - Electronic Digital Balance (6000g) - Type: (EWJ6000 - 1N) - Serial No.: (WF14199926) - Calibration Certificate No.: (1938/13 C008/170/1953/2025).
  - Measuring Tape - Type (BAKR)- Calibration Certificate No.: (107/63 C041/14/100/2026).
- Test Sample:
  - Test sample was choosing under the responsibility of the client.
- Tests:
  1. Surface condition.
  2. Inertness.
  3. Wrapping test for aluminum wires.
  4. Wrapping test for zinc coating.

F - 07 - 08 – 01- 02 Cairo / Alexandria Desert Road - Km 27 - Tel: 35390731 - Fax: 35390728 Page 1 of 8

Ministry of Electricity and Renewable Energy
Egyptian Electricity Holding Company
Laboratories, research and testing sector
Extra High Voltage Research Center
Test Report No. (568/2026)

وزارة الكهرباء والطاقة المتجددة
الشركة القابضة لكهرباء مصر
قطاع المعامل والبحوث والاختبارات
مركز أبحاث الجهد الفائق
تقرير رقم (٥٦٨/٢٠٢٦)

TEST REPORT

REPORT No. (568/2026)

- **CLIENT** : AL- TWAKOL INDUSTRIAL INVESTMENT
  31A 26 July St., Azbakya , Cairo
- **Report Date:** 29/9/2026
- **Place:**
  - **Extra High Voltage Research Center Laboratories.**
  - Internal code: TO - AC - 26 - 09 - 14 - 03
- **Requirements:**
  - Type tests according to IEC standard
- **Standard Specifications:**
  - IEC (61089) / (1991) :Round wire concentric lay over head electrical standard conductors.
  - IEC (62641) / (2022) :Conductors for overhead lines - Aluminium and aluminum alloy wire concentric lay stranded conductores
  - IEC (63248) / (2022) :Conductors for overhead lines – Coated or cladded or metallic wire for concentric lay stranded conductores
  - EDMS(04-100-1)/(2020):Specifications for overhead Aluminium Conductors Steel Reinforced (ACSR).
  - BS EN (50182)/(2001) :Conductors for overhead lines - Round wier concentric lay stranded conductores
  - BS EN (50189) / (2000) :Conductors for overhead lines - zince coated steel wires
- **Description of Specimen :**
  - Aluminum Conductor Steel Reinforced (ACSR) - Size: (380/50) mm^2.
  - **Specimen Receipt Date:** 14 / 9 / 2026
- **Description of Test Equipment:**
  - Testing machine (25) kN - Model: (25ST) - Serial No.: (ST-AFR-02631GB) - Calibration certificate No.: (5330/14C002/60/174/2025).
  - Testing machine (500) kN - Model: (HB 315) - Serial No.: (3618) - Calibration certificate No.: (246/2026/3) - (Witness)
  - Winding Resistance - Type (WR50-12) - Serial No.: (303-174) - certificate No.: (4524/21C026/41/546/2025).
  - Electronic Digital Balance (6000g) – Type: (EWJ6000 - 1N) - Serial No.: (WF1419926) - Calibration Certificate No.: (1938/13 C008/170/1953/2025).
  - Measuring Tape - Type (BAKR)- Calibration Certificate No.: (107/63 C041/14/100/2026).
- **Test Sample:**
  - Test sample was choosing under the responsibility of the client.
- **Tests:**
  1. Surface condition.
  2. Inertness.
  3. Wrapping test for aluminum wires.
  4. Wrapping test for zinc coating.

Ministry of Electricity and Renewable Energy
Egyptian Electricity Holding Company
Laboratories, research and testing sector
Extra High Voltage Research Center
Test Report No. (568/2026)

وزارة الكهرباء والطاقة المتجددة
الشركة القابضة لكهرباء مصر
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مركز أبحاث الجهد الفائق
تقرير رقم (٥٦٨/٢٠٢٦)

5. Design and construction requirements.
6. Determination of direction lay.
7. Determination of lay ratio.
8. Resistivity test.
9. Mass per unit length.
10. Determination of mass of zinc coating.
11. Zinc dip test.
12. Adhesion of zinc coating
13. Tensile test for wires.
14. Elongation test for ZINC coated steel wires.
15. Stress at 1% extension.
16. Tensile test of the conductor.
17. Joints in aluminum wires

- **Test Method and Results:**

1. **Surface condition:**

Testing Date: 14/9/2026 **Testing Engineer**: Usama Elseify

- The test was carried in accordance with clause (6.4.1) of BS EN (50182) and clause (6.6.5) of IEC (61089).
- The test result of the surface condition test following table:

Item	Unit	Requirement	Result
Surface condition	Visual	The surface of the conductor shall be free from all imperfections to the eye (normal corrective lenses accepted), such as nicks , indentations , etc., not consistent with good commercial practice.	Pass

- **The ACSR conductor complied with the requirements.**

2. **Inertness:**

Testing Date: 14/9/2026 **Testing Engineer**: Mohamed Sabry

- The test was carried in accordance with clause (6.4.3) of BS EN (50182).
- The test result of the inertness test is shown in the following table:

Item	Unit	Requirement	Result
Inertness	Visual	All steel wires shall lie naturally in their position in the standard core , and where the core is cut, the wire ends shall remain in position or be readily replaced by hand and then remain approximately in position . this requirement also applies to the aluminum wires of a conductor.	Pass

- **The ACSR conductor complied with the requirements.**

3. **Wrapping test for aluminum wires:**

Testing Date: 14/9/2026 **Testing Engineer**: Mohamed Sabry

- The test was carried in accordance with clause (6.5.2) of BS EN (50182) and clause (6.4.5) of IEC (62641).
- The test result of the wrapping test for aluminum wire is shown in the following table:

Item	Unit	Requirement	Result
Wrapping test	Visual	No break	Pass

- **The ASCR conductor complied with the requirements.**

4. Wrapping test for zinc coating:

Testing Date: 14/9/2026 **Testing Engineer** Mohamed Sabry

- The test was carried in accordance with clause (6.5.2) of BS EN (50182), clause (7.4.4.2) of IEC(63248) and table A.4 (group 1,ST1A).
- The test result of the wrapping test for zinc coating test is shown in the following table:

Item	Unit	Requirement	Result
Wrapping test	Visual	No break	Pass

- **The ACSR conductor complied with the requirements.**

5. Design and Construction Requirements :

Testing Date: 14/9/2026 **Testing Engineer**: Usama Elseify

- The design and construction requirements have been measured according to clause (4) of EDMS (04-100-1), clauses (6.6.1&6.6.2) of IEC (61089), Table (2) of IEC (62641) and Table (A.4) of IEC (63248) .
- The measured values are shown in the following table:

Item	Unit	Requirement	Measured / determined
- Overall conductor diameter	mm	27	27.09
- Diameter Aluminum wires	mm	3	3.01
- No of Aluminum wires	---	54	54
- Diameter of steel wires	mm	3	3
- No of steel wires	---	7	7

- **Tolerance of conductor diameter: ±1%.**
- **Tolerance of Aluminum wires diameter: ±1%.**
- **Tolerance of Steel wires diameter: ±0.05 mm.**

- **The ACSR conductor complied with the requirements.**

6. Determination of direction lay :

Testing Date: 14/9/2026 **Testing Engineer:** Usama Elseify

- The direction of lay of the conductor was measured in accordance with clause (5.4.2) of IEC (61089) and clause (6.4.4) of BS EN (50182).
- The lay direction is shown in the following table:

Item	Lay direction	
	Requirement	Determined
- Direction of lay for conductor outer surface	Right-hand	Right-hand

- **The ACSR conductor complied with the requirements.**

7. Determination of lay ratio:

Testing Date: 14/9/2026 **Testing Engineer**: Mohamed Sabry

- The lay ratio of the conductor was measured in accordance with clause (5.4.5) of IEC (61089) and clause (6.4.4) of BS EN (50182).

M.Sabry Usama Elseify

Ministry of Electricity and Renewable Energy
Egyptian Electricity Holding Company
Laboratories, research and testing sector
Extra High Voltage Research Center
Test Report No. (568/2026)

وزارة الكهرباء والطاقة المتجددة
الشركة القابضة لكهرباء مصر
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مركز أبحاث الجهد الفائق
تقرير رقم (٢٠٢٦/٥٦٨)

- The measured value is shown in the following table:

Item	Lay ratio (%)	
	Requirement	Measured
- Lay ratio for conductor outer surface	10 - 14	12.66

- **The ACSR conductor complied with the requirements.**

8. Resistivity test:

Testing Date: 15/9/2026 **Testing Engineer:** Mohamed Sabry

- The electrical resistivity was measured for Aluminum wire in accordance with clause clause (6.4.7) and table (1) of IEC (62641).
- The measured value is shown in the following table :

Test	Requirement	Measured
- Electrical resistivity for Al. wires at 20 °C ($\Omega.mm^2/m$)	≤ 0.028264	0.027869

- **The ACSR conductor complied with the requirements.**

9. Determination of mass of zinc coating:

Testing Date: 15/9/2026 **Testing Engineer:** Mohamed Sabry

- The mass per unit length was measured for the conductor in accordance with clause (6.6.3) of IEC (61089)), EDMS (04-100-1) and clause (6.4.6) of BS EN (50182).
- The measured value is shown in the following table :

Test	Requirement	Measured
- Mass of zinc coating (gm/m^2)	≥ 245	269.3

- **The ACSR conductor complied with the requirements.**

10. Mass per unit length.

Testing Date: 15/9/2026 **Testing Engineer:** Usama Elseify

- The mass per unit length was measured for the conductor in accordance with clause (6.6.3) of IEC (61089)), EDMS (04-100-1) and clause (6.4.6) of BS EN (50182).
- The measured value is shown in the following table :

Item	Requirement	Determined
- Mass per unit length (kg/km)	1442.5±2%	1458.1

- **The ACSR conductor complied with the requirements.**

11. Zinc dip test:

Testing Date: 15/9/2026 **Testing Engineer** Mohamed Sabry

- The test was carried in accordance with clause (6.5.2) of BS EN (50182) and clause (7.4.5.3) and table (A.12) class (A) of IEC (63248).

M.Sabry Usama Elseify

- The test result of the zinc dip test is shown in the following table:

Item	Unit	Requirement	Result
Dip test	Visual	No correlation between the number of dips and the mass of coating	Pass

- **The ACSR conductor complied with the requirements.**

12. Adhesion of zinc coating

Testing Date: 15/9/2026 **Testing Engineer:** Mohamed Sabry

- Test was carried in accordance with clause (6.5.2) of BS EN (50182) , clause (11.7) of BS EN(50189) and clause (7.4.5.2.1) of IEC (63248) .
- The test result of the adhesion of zinc coating is shown in the following table:

Item	Unit	Requirement	Result
Adhesion test	Visual	The zinc coating shall remain firmaly adhered to the steel and shall no crack nor flack	Pass

- **The conductor complied with the requirements.**

13. Tensile test for wires:

Testing Date: 16/9/2026 **Testing Engineer:** Mohamed Sabry

- The mechanical tensile strength was measured on wires in accordance with clause (7.4.3.2) of IEC (63248), clause (6.5) of IEC (61089) ,clause (6.4.3) of IEC (62641) and clause (6.5.2) of BS EN (50182).
- The measured values are shown in the following table:

No.	Diameter (mm)	Cross section area (mm^2)	Tensile strength (M.Pa)	
			Requirement	Result
Aluminum wires				
1	3	7.06	≥ 165	187.15
2	3.01	7.11		188.2
3	3.01	7.11		189.17
4	3.01	7.11		197.63
5	3.01	7.11		183.05
6	3.01	7.11		173.49
7	3.01	7.11		196.7
8	3.01	7.11		187.41
9	3	7.06		194.2
10	3	7.06		183.71
11	3	7.06		191.16
12	3	7.06		186.69
13	3.01	7.11		181.19
14	3.01	7.11		189.47
15	3	7.06		195.07
16	3	7.06		179.63
17	3.01	7.11		188.5
18	3.01	7.11		192.35
19	3.01	7.11		186.3
20	3.01	7.11		177.19
21	3.01	7.11		190.03
22	3.01	7.11		179.7
23	3.01	7.11		191.12
24	3	7.06		175.3
25	3.01	7.11		181.02
26	3	7.06		179.6
27	3	7.06		188.14
28	3	7.06		195.2
29	3	7.06		178.9
30	3	7.06		181.08
31	3.01	7.11		185.44
32	3.01	7.11		193.7
33	3	7.06		181.22
34	3	7.06		195.06
35	3	7.06		177.79
36	3	7.06		183.2
37	3.01	7.11		197.58
38	3.01	7.11		182.7
39	3.01	7.11		190.05
40	3.01	7.11		182.2
41	3.01	7.11		177.38
42	3	7.06		181.07
43	3.01	7.11		182.31
44	3.01	7.11		178.1
45	3.01	7.11		189.5
46	3.01	7.11		173.31
47	3	7.06		177.9
48	3.01	7.11		181.66
49	3.01	7.11		175.3
50	3.01	7.11		183.04
51	3	7.06		193.7
52	3	7.06		188.13
53	3	7.06		187.85
54	3.01	7.11		176.6
Steel wires:				
1	3.01	7.11	≥ 1300	1381.3
2	3	7.06		1347.5
3	3.01	7.11		1380
4	3	7.06		1344.8
5	3	7.06		1365.02
6	3	7.06		1377.6
7	3	7.06		1342.9

- **The ACSR conductor complied with the requirements.**

14. Elongation test for ZINC coated steel wires:

Testing Date: 16/9/2026 **Testing Engineer:** Mohamed Sabry

- The test was carried in accordance with clause (6.5.2) of BS EN (50182) and clause (7.4.3) IEC (63248).
- The result of the elongation test for zinc coated steel wire is shown in the following table:

Item	Unit	Requirement	Result
Elongation test	%	3.5	7.09

- **The ACSR conductor complied with the requirements.**

15. Stress at 1% extension:

Testing Date: 16/9/2026 **Testing Engineer:** Mohamed Sabry

- The test was carried in accordance with standard clause (6.5.2) of BS EN (50182) and clause (7.4.3.2) of IEC (63248).
- The test result of the stress at 1% extension is shown in the following table:

Item	Unit	Requirement	Result
Stress at 1% extension	Mpa	1100	1241

- **The ACSR conductor complied with the requirements.**

16. Tensile test of the conductor:

Testing Date: 23/9/2025 Witnees **Engineer:** Usama Elseify

- Tensile test of the conductor is carried out according to clause (6.5.3) of IEC (61089).
- The test has been performed on the one (1) specimen.
- The breaking strength of conductors was determined by pulling a conductor in atensile testing machine. For the purposes of this test, appropriate fittings was installed on the ends of the conductor samples.
- The breaking strength of the conductor shall be determined by the load attained at which one or more wires of the conductor are fractured.
- The test sample was held in a tensile testing machine and the load gradually increased until it reaches the specified minimum failure load (115.23) kN (95% of the Rated Tensile Strength [RTS = 121.3kN] according to EDMS (04-100-1). This load kept constant for (60 s).
- Regarding failure load, the test is passed if failure of the conductor does not occur during the test.
- **The conductor passed the test without failure occurs.**

M. Sabry Usama Elseify

Ministry of Electricity and Renewable Energy
Egyptian Electricity Holding Company
Laboratories, research and testing sector
Extra High Voltage Research Center
Test Report No. (568/2026)

وزارة الكهرباء والطاقة المتجددة
الشركة القابضة لكهرباء مصر
قطاع المعامل والبحوث والاختبارات
مركز أبحاث الجهد الفائق
تقرير رقم (٢٠٢٦/٥٦٨)

17- Joints in aluminum wires:

Testing Date: 24/9/2026 **Testing Engineer:** Usama Elseify

- The joints in Aluminum wires test is carried out according to clause (6.5.4) of IEC (61089).
- The manufacturer provided the joint in one stranded of the outer layer of the conductor by using cold pressure buttwelded method.
- The conductor passed the test if its withstand a stress equal (130) Mpa without the fracture of the welded wire .
- The collapse of welded wire occurred at a value of (138.3) Mpa.

▪ Conclusion:

- **The Aluminum Conductor Steel Reinforced (ACSR) - Size (380/50) mm² - Manufactured by AL- TWAKOL INDUSTRIAL INVESTMENT, achieved the requirements of tests mentioned in this report according to IEC(61089), IEC(63248), BS EN (50182), BS EN (50189) , IEC (62641) and EDMS (04-100-1).**

▪ Notes:

- Tests were carried out on the above specimen only without any responsibility concerning other untested specimens
- The tests were carried out without any obligation on Egyptian Electricity Holding Company.
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▪ Test Engineers:

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29/9/2026

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Sherif ...

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